Intercity Yb/Sr optical clock comparison at 2.3×10^{-18}

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Abstract

Redefining the International System of Units (SI) second requires independent frequency-ratio measurements between different types of optical clocks in different laboratories, with an overall measurement uncertainty below 5×10^{-18} and consistent results. However, the self-assessed uncertainties of these clocks have not yet been validated by independent frequency-ratio comparisons. Here we compare a ytterbium optical lattice clock in Wuhan with a strontium optical lattice clock in Shanghai—approximately 700 km away—using a phase-stabilized fiber link spanning 1 350 km. The Yb/Sr frequency ratio is determined as 1.207 507 039 343 337 720 1(28), with a Bayesian statistical uncertainty of 1.14×10^{-18} and a total uncertainty of 2.34×10^{-18}. The value differs from that of NIST–JILA 1.207 507 039 343 337 723 0(37) by -2.87×10^{-18}, consistent within the margin of statistical error. To our knowledge, this is the first intercity, cross-species frequency-ratio measurement to meet the uncertainty requirements outlined in the roadmap for redefining the SI second, and together with the NIST–JILA result it satisfies the mandatory criteria (I.2.b) for the redefinition, confirming the validity of uncertainty assessments for both ytterbium and strontium optical clocks.

1 Introduction

Optical lattice and ion clocks now reach fractional systematic uncertainties at or below 2×10^{-18} [1–10], two orders of magnitude better than current primary frequency standards [11;12]. Comparing independent clocks in different regions and laboratories verifies these self-assessed precision budgets more objectively: the environmental conditions and geodetic assumptions entering a self-evaluation are typically tied to a single location, and identical methods within one laboratory may share common systematic biases. Linking clocks of different types and locations into a collaborative network is therefore a key pathway toward establishing optical clocks as a verifiable metrological infrastructure [13;14]. The idea has moved from proposal to practice: fiber links with noise levels below those of the clocks themselves now span thousands of kilometers [15;16], and multi-clock campaigns have measured tens of ratios among clocks in six countries [17] and across a continental network [18]. Satellite-based advanced links with comparable performance [19–21] are also being developed and will interconnect with fiber networks to form a future global optical clock network.

The most immediate purpose of such a network is the redefinition of the SI second. One of the mandatory requirements (I.2.b) in the roadmap for this redefinition [22] is that frequency ratios of different clock types be measured at least twice by different institutes in agreement with an overall uncertainty of the comparison $\Delta\nu/\nu < 5 \times 10^{-18}$—a requirement reiterated in the 2025 CCTF progress report [23;24]. However, comparisons across independent institutions and different locations have reached at best a precision of 7.7×10^{-18} [18]. So far, no frequency-ratio measurements between clocks in different cities have satisfied this requirement [25].

Closing that gap matters beyond the definition itself. Chronometrically leveled clocks map the gravitational potential of the Earth at centimeter resolution [1;26;27], and clock networks have been proposed and used as sensors for dark matter [28–30]. All of these applications assume that remote clocks agree at a level that, until now, has been demonstrated only locally, in two experiments: a Sr/Yb/Al^+ triple-clock ratio measurement with an uncertainty of 3.2×10^{-18} by NIST–JILA



Figure 1: The two-clock network. An IAPMST ytterbium clock in Wuhan and a USTC strontium clock in Shanghai are connected by a phase-stabilized 1 550 nm fiber link. Relay stations along the route carry erbium-doped fiber amplifiers and fiber-noise-cancellation loops.

in Boulder (US)[31], and a Yb^+/Sr measurement with 4.3×10^{-18} by PTB in Braunschweig (Germany)[32].

Here we report an intercity, cross-species frequency-ratio measurement between an IAPMST ytterbium lattice clock in Wuhan and a USTC strontium lattice clock in Shanghai, about 700 km apart, connected by a 1 350 km fiber link. We determine the Yb/Sr ratio with a total uncertainty of 2.34×10^{-18}; gravitational redshift effects have been corrected, whereas measurement error and the time-varying gravitational redshift term associated with gravitational tides have been retained as uncertainty components. The result establishes the two sites as a prototype node pair for a distributed optical-clock network, and completes, at intercity scale, the class of ratio measurements that the redefinition process depends on.

2 Results

2.1 Network and observation campaign

Figure 1 shows the two-clock network. A ytterbium (Yb) lattice clock at IAPMST[8] in Wuhan (systematic uncertainty 1.3×10^{-18}) and a strontium (Sr) lattice clock at USTC[5] in Shanghai (0.92×10^{-18}) are separated by 700 km and compared through a 1 350 km phase-stabilized fiber link (2 700 km loop-back) on a commercial telecom network used non-exclusively, with the carrier transmitted from Shanghai to Wuhan on the ITU-T 100 GHz DWDM channel C34 (1 550.12 nm, 193.4 THz). The link extends the field-deployed link of ref.[16], using the same cascaded fiber-noise-cancellation technique; its added instability stays below the clock noise for averaging times beyond 100 s.

Both combs are self-referenced and phase-locked to the local clock signal, and all other frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations (698/1397 nm on the Sr side; 578/1156 nm on the Yb side), so the two clocks are the system's only independent frequency sources. The recorded quantity is the out-of-loop beat between the received carrier and the local Yb-referenced comb, sampled every second and converted to fractional units by normalizing to the carrier frequency f_{1550} (1 550.12 nm); the fiber loop-back is monitored out of loop throughout. The geopotential difference between the two clocks

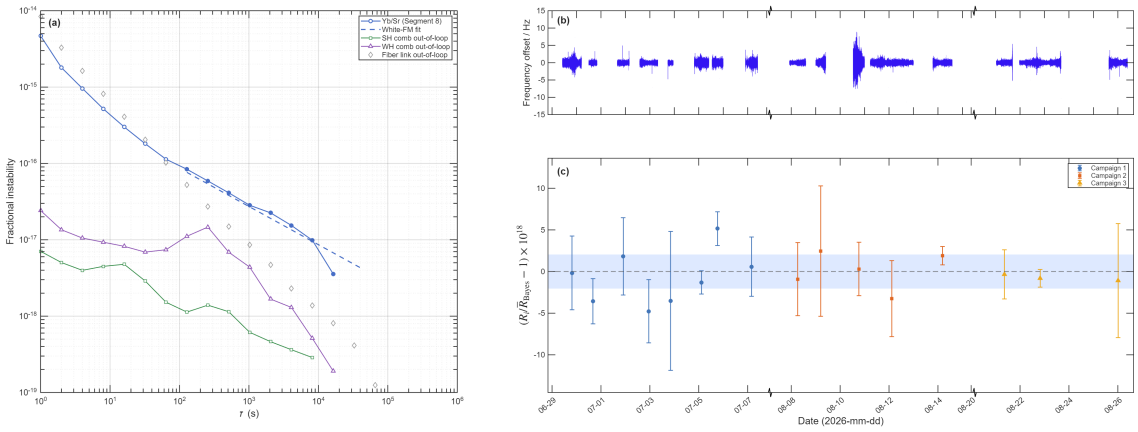


Figure 2: Frequency-ratio measurement results. (a) Fractional instability of a representative segment (Yb/Sr, segment 8); the points are the overlapping Allan deviation at different averaging times τ. The solid line is a white-frequency-noise model $\sigma_y(\tau) = \sigma_0/\sqrt{\tau/\mathrm{s}}$; the statistical uncertainty is the last computable point of the stability record, without extrapolation. The gray markers are the out-of-loop instabilities of the two frequency combs (SH, WH) and of the fiber link. (b) Per-segment ratio deviations from the final reported value, with error bars from the procedure in (a). (c) The same deviations by campaign date.

was measured by spirit leveling to be $\Delta W = 280.042 \pm 0.085\ \mathrm{m^2\,s^{-2}}$ (see Methods). Each retained segment yields a frequency-ratio estimate through the fixed frequency-chain relations (see Methods).

2.2 Clock-ratio determination

Over two months (58 days) the two clocks and their transfer chains ran synchronously for a total of 274 effective hours, recorded as 16 jump-free segments. The results are presented in Figure 2. In panel (a), the overlapping Allan deviation of the fractional ratio for a representative segment follows a white frequency noise law over most of the record. The statistical uncertainty of each segment is taken from the last computable point of its stability record, rather than by extrapolating the white-noise fit to the segment duration; together with the segment mean, this provides one data point per segment (Figure 2b,c).

We determine the Yb/Sr frequency ratio from the 16 segments. Each segment yields a ratio estimate with its own statistical uncertainty (Methods). The segment estimates are consistent with their individual uncertainties, as quantified by the combination statistics in Table 1. Following recent ratio comparisons[18;31], we combine the segments with a random-effect model that admits a common excess scatter[33;34], and evaluate three estimators (Table 1). The three agree at the 10^{-18} level; the Bayesian random-effect estimate, which carries the largest standard uncertainty, is reported as the primary value,

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1, \tag{1}$$

with statistical uncertainty 1.139×10^{-18}. The five segments flagged as model-limited in the per-segment table (Methods) are retained with their measured uncertainties.

2.3 Uncertainty budget

Beyond the statistical uncertainty of the ratio combination, the comparison carries several systematic contributions: the published evaluations of the two clocks, the geodetic determination of

Table 1: Statistical combination of the 16 segment ratios. WLS denotes weighted least squares with inverse-variance weights $w_i = 1/u_i^2$; ξ is the inferred common excess scatter. The weighted combination yields $\chi^2_{\mathrm{red}} = 0.412$ for 15 degrees of freedom ($p = 9.8 \times 10^{-1}$) and a Birge ratio of 0.642. All three estimators agree at the 10^{-18} level.

Estimator	Yb/Sr	u (10^{-18})	ξ (10^{-18})
WLS (inverse variance)	1.207 507 039 343 337 720 130	1.128	—
Mandel–Paule	1.207 507 039 343 337 720 130	1.128	0.00
Bayesian random effect	1.207 507 039 343 337 720 134	1.139	0.14

Table 2: Uncertainty budget of the Yb/Sr frequency-ratio measurement. Components are treated as independent.

Component	Contribution (10^{-18})
Statistical (Bayesian random effect)	1.139
Sr systematics	0.92
Yb systematics	1.3
Static gravitational potential	0.946
Fiber link	0.1
Frequency comb	0.1
Time-varying redshift	0.85
Total (quadrature)	2.34

the gravitational-potential difference between the sites, and the residuals of the transfer subsystems. We discuss each in turn and combine them into the total standard uncertainty (Table 2).

The Sr clock contributes 0.92×10^{-18} from its published systematic evaluation[5], and the Yb clock 1.3×10^{-18}[8]. The static gravitational-potential term is the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$ converted to fractional frequency as $u(\Delta W)/c^2$, giving 0.946×10^{-18}; this is the same leveled geopotential difference already applied as the static correction (Section 2.1). The frequency conversion stages each contribute 0.1×10^{-18} and 0.1×10^{-18} from the frequency-reference settings of the frequency conversion.

The time-varying gravitational redshift is more involved. Solid Earth and ocean tides, together with atmospheric, hydrological and non-tidal ocean loading, move the local potential by amounts reaching the 10^{-18} level over the campaign[35–37], and no single tide model accounts for all of them consistently: correcting the beat record with the supplied tide series degrades rather than improves the fit, and a second, independent tide model does the same while moving the center in the opposite direction (Methods, Table 4). The sign disagreement between two independent models confirms that the templates are incomplete at this level, so the time-varying term is carried as an uncertainty rather than a correction, at the larger of the two model shifts, 0.85×10^{-18} (Methods). A physically complete, data-consistent treatment of time-dependent gravity is deferred to future work.

Combining the statistical and systematic components in quadrature gives the total standard uncertainty 2.34×10^{-18}; the reported ratio is therefore

$$\mathrm{Yb/Sr} = 1.207\,507\,039\,343\,337\,720\,1(28), \tag{2}$$

where the parenthesized digits are the total standard uncertainty in the last two decimal places of the value.

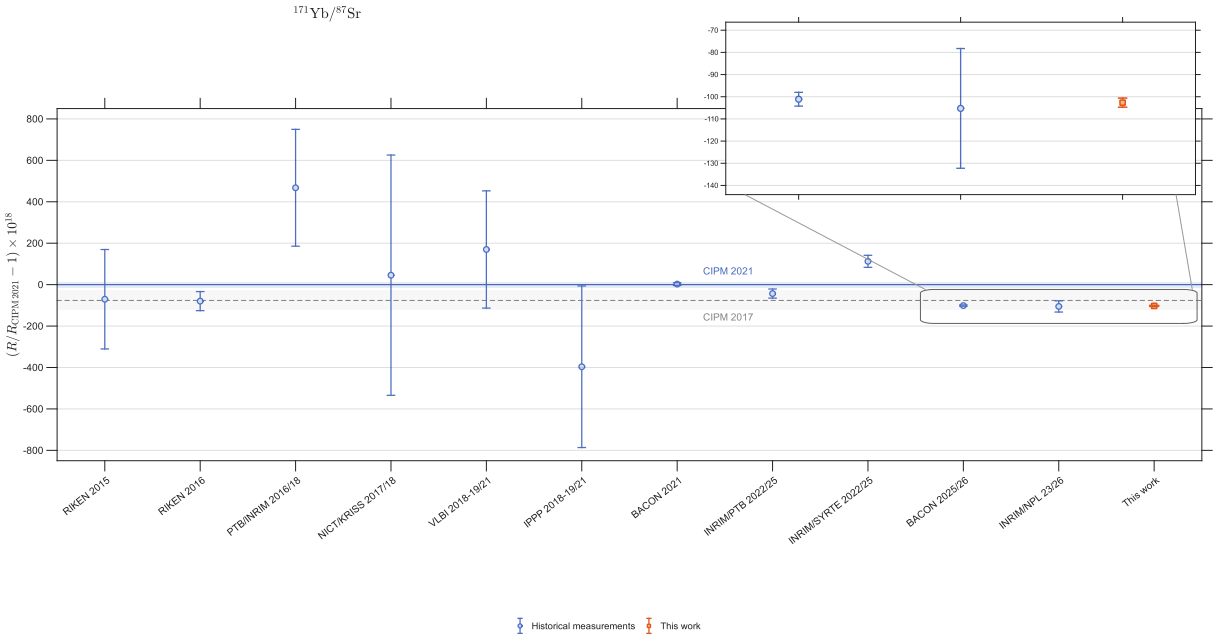


Figure 3: Evolution of the Yb/Sr frequency-ratio measurements. Deviations from the final value are shown against the measurement year for the historical determinations and for this work. The successive measurements converge toward agreement at the few-10^{-18} level.

3 Discussion and outlook

The value reported here agrees with the two existing measurements of the Yb/Sr ratio at the level set by the redefinition roadmap (Figure 3): relative to the NIST–JILA value 1.207 507 039 343 337 723 0(37)[31] our central value differs by -2.87×10^{-18}, and relative to the European-network value 1.207 507 039 343 337 718(32 by $+2.13\times10^{-18}$, both well within the 5×10^{-18} agreement threshold of the roadmap's frequency-ratio criterion[22;24]. To our knowledge, this is the first intercity, cross-species frequency-ratio measurement meeting that requirement (as of October 2026).

The three independent Yb/Sr values also track the progress of the comparison technology itself: from the first measurement in 2014[38] and its refinement to 5×10^{-17}[39], through the BACON campus network[30] and the European fiber network[18], to 3.2×10^{-18} on a single campus[31]. Our measurement closes this sequence at intercity scale with 2.34×10^{-18}, and the three latest, independently obtained values agree at the few-10^{-18} level. The convergence works in both directions: each comparison validates the others, and the agreement feeds back into the evaluation of the methods and error budgets of the individual laboratories—what the redefinition needs is no longer a single laboratory's claim about its own clock, but the agreement between independent measurements.

The final uncertainty is comparable to, and slightly below, that of the NIST–JILA campus comparison, even though the present measurement operates on a distributed system that carries additional uncertainty from the fiber link, the two frequency combs and the geodetic potential difference. The compensating factor is time: the 274-hour campaign record is considerably longer than a typical single-campus measurement window, and longer continuous averaging is the direct route to a smaller statistical term, provided the record remains jump-free. Keeping the record clean was itself a technical achievement of the campaign, made possible by the jump-suppression and monitoring controls on the fiber link; the resulting segments carry almost no rejected samples.

Changes in the Earth's gravitational potential enter the comparison directly. The static difference between the two clock transitions is fixed by first-order spirit leveling and applied as a correction; of the time-varying contributions, the tidal term is already appreciable at our precision and is therefore carried as a data-driven uncertainty item rather than corrected (Section 2.3),

while co-seismic steps from the regional seismicity during the campaign fall well below the reported precision. The effect grows with the scale of the comparison: on longer, continental- and intercontinental-scale baselines the time-varying redshift becomes one of the dominant systematic terms[37]. That same sensitivity turns the network into an instrument for the Earth: clock comparisons over large baselines directly sense the time-variable gravitational potential, providing an independent means of testing and calibrating tide and loading models at the 10^{-18} level—the chronometric-geodesy applications proposed for optical-clock networks[1;26;27]. Conversely, tidal knowledge gained in this way feeds back into the comparisons themselves: with data-consistent models, the time-varying term can be corrected rather than carried, extending the accuracy reach of large-scale clock comparisons. A fully self-consistent treatment of the time-dependent potential would require a dedicated geodetic campaign with in-situ gravimetry and site-specific loading models and is left to future work.

The campaign also demonstrates the operational character that a future optical-clock network will need. The reproducibility of the reported ratio rests on more than the retained record: before the comparison, the two clocks were deliberately switched off and restarted to test their restart reproducibility, and the laboratory environments were kept under tight control, so that the clocks' ability to return to the same frequency is itself a tested property of the measurement. Over the two-month campaign the two clocks were then restarted, the transfer chain re-locked, and the beat record interrupted many times; the segment-level ratio estimates show no evidence of a systematic offset between the three measuring stages, so the reproducibility of the comparison does not depend on continuous operation of either system. This is the practical counterpart of the coordinated multi-clock campaigns[17;18]: the network tolerates interruption, provided each retained window is self-consistently characterized.

Several limitations bound the present result. First, the per-segment statistical uncertainties are the dominant budget term, and five of the sixteen segments carry model-limited slope diagnostics; the random-effect treatment accounts for any remaining excess scatter at the level of the reported uncertainty, but a longer record would reduce the statistical term. Second, the systematic evaluations of the two clocks are those published by the laboratories; independent verification of those budgets, as opposed to the transfer, is not attempted here. Third, the geopotential difference rests on a single first-order leveling campaign: the available GNSS-plus-geoid-model calibration is too imprecise (Methods) to act as an effective cross-check, and closing this gap will require transportable optical clocks with higher frequency reproducibility, rather than geodetic techniques alone; the provenance of the supplied tide series and the exact accounting of the full sample totals remain external items pending confirmation. Finally, the present campaign meets the uncertainty requirement of the roadmap but not the full criterion, which additionally requires each ratio to be measured at least twice by different institutes and repeated over an extended period.

Looking ahead, the result establishes this link as a prototype node pair of a distributed optical-clock network: the same infrastructure can be extended to further nodes, and the technique demonstrated here (clocks at different sites agreeing at the redefinition threshold) is the operation that both the redefinition and post-redefinition applications require. The building blocks for such a network are now in place: terrestrial fiber links of the kind used here[15;16] carry comparisons over thousands of kilometers within continents, while free-space and satellite-based links are closing the remaining gaps[19–21;40]. The natural next step is a hybrid architecture in which optical-clock nodes on continental fiber backbones are interconnected across oceans by satellite or free-space optical links, so that any pair of clocks on Earth can be compared at the level demonstrated here. Realizing it will put the same demands on geodetic infrastructure that this comparison faced[1;26].

4 Conclusion

We have compared an ytterbium lattice clock in Wuhan and a strontium lattice clock in Shanghai over a 1 350 km fiber link and determined the Yb/Sr frequency ratio with a total standard uncertainty of 2.34×10^{-18}. This is, to our knowledge, the first intercity, cross-species ratio measurement to meet the uncertainty requirement of the roadmap for the redefinition of the SI second, and it brings the three independent Yb/Sr determinations into agreement at the few-10^{-18} level. The remaining limitations are the systematic evaluations of the two clocks and the geodetic treatment of the gravitational potential, whose time-varying component is not yet correctable with available tide models. With these addressed, the two sites form a working node pair that lays the foundation for extending the network further: joining more optical-clock nodes through terrestrial fiber backbones and interconnecting them across continents by satellite and free-space links will turn the comparison demonstrated here into a global optical-clock network. Such a network will provide the verifiable metrological infrastructure that the redefinition of the SI second requires and, beyond it, will support chronometric leveling of the gravitational potential at centimeter resolution and distributed searches for new physics with clock arrays.

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Methods

Leveling determination of the geopotential difference

The static gravitational-redshift correction is fixed by the geopotential difference between the atomic-transition positions of the two clocks, determined by a first-order spirit leveling campaign carried out specifically for this comparison. The leveling line, about 1 113 km long, runs between the Wuhan benchmark WS-03 (at IAPMST) and the Shanghai benchmark WS-01 (at USTC) through Ezhou, Huangshi, Huanggang, Lu'an, Hefei, Wuhu, Guangde and Suzhou. Digital levels were used with bar-code invar staffs, with sight lengths not exceeding 30 m; the line was observed in 6 km to 8 km sections in both directions, each checked in the field. Gravity was observed along the line at the leveling points with two relative gravimeters. The orthometric height difference between the benchmarks follows from the leveled normal heights after the normal-to-orthometric conversion, and the geopotential difference is integrated section by section, $\Delta W_{AB} = -\sum_i g_i \, \Delta H_i$, where g_i is the mean gravity and ΔH_i the orthometric height difference of the i-th section. Combining this with the two local offsets between benchmark and clock platform (obtained by trigonometric leveling) and the platform-to-transition offsets from the clock design parameters gives the atomic-transition geopotential difference $\Delta W = 280.042 \pm 0.085 \ \mathrm{m^2\,s^{-2}}$, equivalent to an optical fractional shift of -3.116×10^{-15}. The accidental error per kilometer of the leveling is $M_\Delta = 0.255 \,\mathrm{mm\,km^{-1}}$, which satisfies the most stringent requirements for first-order leveling in the national standards of China, the United States, Germany and Czechia[41–44].

As an independent check, the geopotential difference was also determined by a combined GNSS and geoid-model calibration: the ellipsoidal heights of the permanent marks were measured by precise point positioning, their vertical offsets to the clock reference surfaces were

leveled locally, and the EGM2008 geoid model was used to convert to a geopotential difference of $280.60 \pm 2.558\ \mathrm{m^2\,s^{-2}}$. The two independent determinations agree within the uncertainties of the GNSS-based value, but the GNSS-plus-geoid calibration is two orders of magnitude less precise than the leveling result, so it cannot act as an effective cross-check at the reported accuracy; a meaningful independent verification will require transportable optical clocks with higher frequency reproducibility.

Experimental setup and frequency transfer

The comparison links two optical lattice clocks at different institutions: an ytterbium (Yb) clock at IAPMST in Wuhan and a strontium (Sr) clock at USTC in Shanghai, about 700 km apart. The two sites are connected by a phase-stabilized fiber link of 1 350 km single-pass length (2 700 km loop-back), operated at 1 550 nm. The route is divided into spans of tens of kilometers, each terminated by a relay station that carries a narrow-linewidth laser together with an erbium-doped fiber amplifier and a fiber-noise-cancellation loop; the added instability of the link is kept below the clock noise level for averaging times beyond 100 s.

Both ends use independent optical frequency combs to bridge the spectral gap between the clock transitions and the telecom carrier. On the Shanghai side the Sr 698 nm transition is transferred via a 1 397 nm continuous laser to the 1 550 nm carrier; on the Wuhan side the received carrier is linked to the Yb 578 nm transition through the 1 156 nm light. All local frequency references and phase-locked loops are derived from the comb repetition rate $f_{\mathrm{rep}} = 200\,\mathrm{MHz}$. The observable is the out-of-loop beat between the received carrier and the local Yb-referenced comb, converted to fractional units with the transfer frequency $f_{1550} = 193.3992\,\mathrm{THz}$. All frequencies in the chain derive from the two clock transitions through fixed factor-of-two relations, $f_{\mathrm{comb}}^{1397} = \frac{1}{2}(1+a)\,f_{\mathrm{Sr}}$ and $f_{\mathrm{comb}}^{1156} = \frac{1}{2}(1+b)\,f_{\mathrm{Yb}}$, where a and b collect the Sr and Yb systematic corrections (density, lattice, second-order Zeeman and blackbody shifts). The static gravitational-redshift correction enters the ratio inversion as a multiplicative factor, $(\mathrm{Yb/Sr})_{\mathrm{corr}} = (\mathrm{Yb/Sr})_{\mathrm{raw}} \times (1+\delta_g)$ with $\delta_g = -3.116 \times 10^{-15}$, derived from the leveled geopotential difference above.

Data processing and segment definition

The counters sample the beat every second in phase-averaging mode. Each accepted segment contributes its longest jump-free record. Table 3 lists the per-segment retained counts, ratio deviations and statistical uncertainties; the category column marks the five segments whose stability slopes fall outside the white-frequency-noise band (model-limited). These five lie at the steep end of a continuous slope distribution, just beyond the acceptance bound, and show no common campaign, weather or laboratory condition that distinguishes them from the remaining segments.

Clock-ratio determination, statistical model and uncertainty budget

Each retained segment yields a clock-ratio estimate R_i from the mean of its retained beat samples, inverted through the fixed frequency-chain relations and corrected for the static gravitational redshift $\delta_g = -3.116 \times 10^{-15}$ derived from the leveled geopotential difference (see above). The per-segment statistical uncertainty u_i is obtained from the overlapping Allan deviation of the beat fractional frequency, evaluated on a tau grid up to a quarter of the longest gap-free run T_i of the segment: u_i adopts the last computable point $u_i = R_i\,\sigma_y(\tau_{\max})$ without extrapolating the stability curve beyond the data. The log-log slope of σ_y against τ is computed as a diagnostic; segments whose slope leaves the white-frequency-noise band are flagged but retained.

The segment estimates are then combined under a random-effect model with a common excess scatter[33;34]: each segment is modeled as $R_i \sim \mathcal{N}(\mu, u_i^2 + \xi^2)$, with ξ added in quadrature to the per-segment uncertainties. The primary value is obtained from a Bayesian random-effect

Table 3: Per-segment data of the 16 segments analyzed in this work. Ratio deviations $y_i = R_i/R_1 - 1$ are relative to the first segment; $u(y_i)$ is the statistical uncertainty of the segment ratio; "Fit" is *est.* for segments whose stability fit lies in the white-frequency-noise band and *lim.* for model-limited fits.

Group	n_{valid}	y_i (10^{-18})	$u(y_i)$ (10^{-18})	Fit
1	68 009	+0.00	4.245	est.
2	29 481	-3.36	3.017	lim.
3	42 891	+2.00	5.880	est.
4	64 280	-4.61	3.674	est.
5	19 532	-3.38	15.250	est.
6	52 140	-1.15	2.230	lim.
7	39 526	+5.31	3.476	lim.
8	43 565	+0.97	9.860	est.
10	57 938	-0.76	5.003	est.
11	35 824	+2.61	5.976	est.
12	40 346	+0.73	4.843	lim.
13	152 994	-3.09	4.019	est.
14	68 618	+1.97	6.254	est.
15	57 649	-0.25	4.028	est.
16	147 640	-0.77	1.809	lim.
17	65 983	-0.89	8.813	est.

model, in which μ and ξ are jointly inferred with flat priors on μ and on $\log \xi$, and the joint posterior $p(\mu, \xi \mid \{R_i, u_i\})$ is marginalized numerically over a 40-decade $\log \xi$ grid; the posterior mean of μ and its posterior standard deviation give the reported central value and statistical uncertainty, with $\xi = 0.14 \times 10^{-18}$. As cross-checks, we also combine the segments by weighted least squares (inverse-variance weights $w_i = 1/u_i^2$), which yields $\chi^2_{\mathrm{red}} = 0.412$ for 15 degrees of freedom ($p = 9.8 \times 10^{-1}$), and by the Mandel–Paule estimator[34], which gives $\xi = 0.00 \times 10^{-18}$; the Birge ratio is 0.642.

The remaining terms of the uncertainty budget (Table 2) are systematic. The Sr and Yb terms are the published evaluations of the two clocks[5;8]; the value adopted for the Yb clock resolves an internal-versus-published discrepancy in favor of the published evaluation. The static-potential term is $u(\Delta W)/c^2$ from the leveling uncertainty 0.085 $\mathrm{m^2\,s^{-2}}$, giving 0.946×10^{-18}. The frequency-conversion stages contribute 0.1×10^{-18} and 0.1×10^{-18} from their frequency-reference settings.

Tidal template and time-varying redshift

The time-varying gravitational-redshift term is quantified from a tidal template. The supplied tide series consists of the solid Earth tide and ocean tide loading displacement difference between the two sites, computed every 30 s with the public software ETideLoad4.5[45], which implements the geophysical models and numerical standards of the IERS Conventions (2010)[46]. The displacement difference is converted to an equivalent geopotential difference with the local gravity g and then to a beat-frequency template through $h = f_{1550}\,\Delta W/c^2$, the same normalization used for the beat record.

To test how well the template fits the data, we repeated the ratio combination with the tide series applied before the inversion, at two fixed response amplitudes: the theoretical amplitude $A = -1$ and the historical amplitude $A = -0.54$ obtained from the demeaned beat waveform in our earlier analysis of the same campaign. The fit degrades rather than improves: the reduced chi-square of the combination rises from 0.41 to 0.57 and 0.54, respectively, and the Bayesian

Table 4: Tidal-template diagnostics for the two independent tide models (16 segments). χ^2_{red} is the reduced chi-square of the ratio combination with the template applied at the historical amplitude $A = -0.54$; the center shift is the Bayesian center change relative to the uncorrected combination, and r is the Pearson correlation of the segment-mean ratio deviations with the segment-mean tidal shift. Both models degrade the fit relative to the uncorrected value ($\chi^2_{\mathrm{red}} = 0.41$) and disagree in the sign of the center shift, so the larger shift is carried as the uncertainty item.

Model	χ^2_{red}	Center shift (10^{-18})	r
ETideLoad4.5	0.54	$+0.85$	$+0.43$
DE421 + FES2022	0.50	-0.21	$+0.40$

center moves by 0.85×10^{-18}. A second, independent tide template was evaluated as a cross-check: the solid tide was computed from the DE421 ephemerides (Earth–Sun–Moon positions) with Love numbers, and the ocean-tide loading correction to the gravitational potential was taken directly from the FES2022 global ocean-tide model, on a 90 s Beijing-time grid. Its fixed-amplitude combination again degrades the fit ($\chi^2_{\mathrm{red}} = 0.50$ at the historical amplitude) but moves the Bayesian center in the opposite direction, by 0.21×10^{-18}. To separate the solid-tide and ocean-loading contributions, we also repeated the segment-mean correlation with the solid tide alone: the Pearson correlation of the ratio deviations with the segment-mean full tide is $+0.43$, which falls to $+0.36$ for the solid tide only, indicating that the dominant coupling comes from the ocean-tide loading term.

Templates that degrade the segment agreement, and whose two independent realizations disagree in sign, are treated as inconsistent with the data: no tidal correction is applied to the reported central value, and the larger of the two center shifts is carried as the time-varying-redshift uncertainty item 0.85×10^{-18} (Table 4).

The components are treated as independent and combined in quadrature, giving the total standard uncertainty 2.34×10^{-18}.

Data availability

The frozen, machine-readable result layer underlying this manuscript is archived in the repository (`https://github.com/hjiang555-a11y/gravitation-whsh`) under `results/audit-v1/verified/`: the manifest with SHA-256 `ac725aaba8ec744bf4e7a17cf37cee41af57461f0d3684584711b24a02b0962e`, the derived ledgers and statistics, and the generated audit report. The raw beat files and the leveled geopotential inputs are not redistributed.

Code availability

The analysis and manuscript pipelines are in the same repository under the Apache License 2.0. The manuscript numbers are generated by `paper/rev2/tools/build_numbers.py`, which reads only the frozen manifest and the curated constants listed in `generated/numbers_manifest.json`.